

# Certificate of Analysis

## PRODUCT DETAILS

|                     |                           |
|---------------------|---------------------------|
| Product Name        | Exosomes (RejuvCORE*)     |
| Batch               |                           |
| Manufacturing Date  |                           |
| Expiry Date         |                           |
| Dosage Form         | Clear-colourless solution |
| Storage Temperature | 4 – 8°C                   |

| TEST                                 | SPECIFICATION                                                                                                | TEST RESULTS                          | REMARKS                                                                               |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------|
| Infectious Diseases Screening        | HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR                                                                     | Not Detected / Non-Reactive           |    |
| Sterility                            | No cloudiness of the test media                                                                              | Negative                              |   |
| Bacterial Endotoxin Test (LAL)       | Endotoxin sensitivity 0.125 EU/mL                                                                            | Negative                              |  |
| Mycoplasma                           | Mean of absorbance <0.05                                                                                     | Negative                              |  |
| Simple Western Blot                  | Positive Marker:<br>Tetraspanin – CD9, CD63<br>Cytosolic – TSG101<br>Negative Marker:<br>Cellular – Calnexin | Detected<br><br>Trace marker detected |  |
| Nanoparticle Tracking Analysis (NTA) | Exosome particle size range<br>30nm – 200nm                                                                  | 10 x 10 <sup>9</sup> particles/mL     |  |

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

\*General Exosomes fortified with Curcumin

**Authorised Signature:**

**Quality Control Department**

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## Supporting Information

### STERILITY TEST <sup>6,8</sup>

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.  
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

### BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST <sup>3,6</sup>

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

### MYCOPLASMA TEST <sup>4,6</sup>

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

### SIMPLE WESTERN BLOT

| Marker | Positive Marker |      | Negative Marker |          |
|--------|-----------------|------|-----------------|----------|
|        | Tetraspanin     |      | Cytosolic       | Cellular |
|        | CD9             | CD63 | TSG101          | Calnexin |
| Result | +               | +    | +               | *        |

'+' indicates the presence of the marker in the cell.  
'\*' indicates the trace of the marker detected in the cell

### NANOPARTICLE TRACKING ANALYSIS (NTA) <sup>7,8</sup>

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



#### REFERENCES

- Chetthimada, Sukrutha, Lorenz, David R., Misra, Vikas, Dillon, Simon T., Reeves, R. Keith, Manickam, Cordelia, 2018, Exosome markers associated with immune activation and oxidative stress in HIV patients on antiretroviral therapy.
- Joana Maia, Silvia Batista, Nuno Couto, Ana C. Gregório, Cristian Bodo, Julia Etzanowska, Maria Carolina Strano Moraes, Bruno Costa-Silva, 2020, Employing Flow Cytometry to Extracellular Vesicles Sample Microvolume Analysis and Quality Control.
- Bacterial Endotoxins. European Pharmacopoeia 5.0; <https://gmp.eu.com/validation/Method/LAL/EUPHARMACPOEIA.pdf>
- US Pharmacopeia <63> Mycoplasma Tests
- ICH Harmonised Tripartite Guideline: Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q6B.
- US Pharmacopeia <71> Sterility Test
- Neha Chopra, Ibrahim Dutt Aray, Namrata Jain, Poonam Yadav, Salma Wajid, Surinder P. Singh, Sangeta Choudhury, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
- Dae Hyun Ha, Hyun-keun Kim, Joon Lee, Hyuck-Hoon Kwon, Gyeong-Hun Park, Seon-Ho Seong Yang, Jee Yoon Jung, Housung Choi, Jun Ho Lee, Sumi Sung, Yong Weon Yi, Byoung Seung Cho, 2020, Mesenchymal Stem/Stromal Cell-Derived Exosomes for Immunomodulatory Therapeutics and Skin Regeneration.

